

Initial Evaluation of the New Real-time Tracking Gradiometer Designed for Small Unmanned Underwater Vehicles

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Abstract- The shallow water localization of buried mines places increased emphasis on sensor and sensor platform size and maneuverability. The Office of Naval Research (ONR) has funded a number of projects to develop efficient buried minehunting capabilities. In particular, they are supporting the development of two Unmanned Underwater Vehicles (UUVs) designed specifically for shallow water minehunting. In addition, the ONR has funded the development of two magnetic sensors for use on these UUV's. This paper documents the recent progress that has been made in operating one of these sensors, the uuv-Real-time Tracking Gradiometer (uuv-RTG) underwater and in motion.

The new uuv-RTG, with a 165 mm baseline, consists of four, 3-axis fluxgate magnetometers, each located within a 3-axis Helmholtz coil. The coils on the fourth magnetometer are not connected to allow its use as a reference sensor that provides magnetic feedback to the other three sensors. The feedback developed by the reference sensor is used to null the signals caused by motion, allowing the three primary sensors to operate at low noise levels in large magnetic fields. These three magnetometers are then used to develop six gradients, five of which are independent. With the appropriate software algorithms, this system is able to provide useful range and bearing to ferromagnetic targets, as well as their magnetic moment.

Initial evaluation of RTG technology underwater was accomplished by installing a field prototype RTG (RTG-fp) with a 305 mm baseline, onboard a towed, low mag sled designed by Florida Atlantic University (FAU) for their Buried Object Scanning Sonar (BOSSII). The RTG-fp operating in unison with the BOSS II system on the BOSS sled, provided limited but useful data regarding its performance and ability to be fused with a sonar, such as the BOSS II.

Prior to their planned integration in late 2005, the new generation uuv-RTG and the BOSS III are currently undergoing performance testing on individual AUVs called Bluefin 12s [1]. Operating a magnetic sensor onboard a small AUVs such as the 324 mm diameter Bluefin 12 is especially difficult. This is due, not only to the close proximity of the magnetometers to multiple magnetic interference sources that are integral to the vehicle, but also to magnetic interference generated by other sensors that are present and share the same payload section. An extensive noise mitigation package consisting of three, 3-axis magnetometers and 2 current sensors were installed on the Bluefin 12 to mitigate this problem.

The uuv-RTG is currently being magnetically characterized. Once the characterization is completed, it will be integrated with BOSS III on a winged version of the Bluefin 12. Here it will be evaluated both independently and in fusion with the BOSS III.

I. INTRODUCTION

Mine localization in shallow water and surf zone environments is very problematic. High probability of target burial and severe environmental noise conditions both deteriorate sensor performance. This place increased demands on the performance of both the sensors and the vehicles that transports them. A single sensor technology is not sufficient due to the wide range of target characteristics and environmental conditions encountered by the system. Therefore, complementary sensors, in a carefully selected suite, are required. The choice of sensors in that suite is strongly dependent on both the targets of interest and environmental conditions. Magnetic sensors have several traits that make them strong candidates for inclusion in any viable mine hunting sensor suite. The environment does not significantly affect magnetic sensors. They are able to detect buried ferrous mines equally as well as proud and can provide important classification information when coupled with complementary technologies. However, because of their inability to effectively detect nonferrous targets and inherent short range, additional complementary technologies such as sonar must always be included.

For many years, vector magnetometers such as fluxgates were considered useful only in a stationary mode or as expensive compasses. This was due to its vector nature and the large signals produced when it was in motion in the Earth's magnetic field. With the advent of improved electronics and unique feedback circuitry, it now becomes possible to use its vector response to an advantage for target localization while on board moving, low-mag platforms such as UUVs.

The requirement to operate in shallow water is driving down the size of not only the transporting platform but its sensor suite as well. The vehicle of choice for shallow water operations is the small UUV. The U.S. Navy is employing them in increasing numbers because of their longer endurance and ability to operate in environments unsuitable for human divers. A special class of UUVs, known as Autonomous Underwater Vehicles (AUVs) can be pre-programmed to perform specific and detailed missions. When outfitted with the proper sensor suite, they are especially suited for searching and localizing on underwater objects of interest. Magnetic sensors, such as the uuv-RTG, must be properly integrated into the AUV so as not to be

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degraded by platform generated Electro-magnetic interference (EMI). This integration is greatly simplified by initially selecting a vehicle with a minimal magnetic signature and which allows maximum sensor isolation from the EMI sources on the platform. Any remaining degradation can then be removed using noise mitigation sensors as discussed in section III, along with custom designed noise reduction software.

The ONR has funded a number of Minehunting projects. In particular, they are supporting the development of two, 324 mm diameter UUVs especially designed for efficient shallow water minehunting. These are the Bluefin 12 and the REMUS 600. In addition, the ONR is funding the development of two magnetic sensors to operate on these UUVs. These sensors are the uuv-RTG which is based on fluxgate technology and presented in this paper, and a more complex, more expensive but longer range sensor called the Laser Scalar Gradiometer (LSG) based on the opto-magnetic properties of helium-4 gas and presented as a second paper in this proceedings [2]. This paper focuses on the mechanical and electrical aspects of integrating the uuv-RTG with the Bluefin 12 (Fig. 1) and summarizes the collected data.

Earlier papers [3-7] specifically discuss candidate magnetic sensor systems, while later papers [8-12] discuss considerations for placing these systems on commercial AUVs. The FAU BOSS III system is also a candidate for inclusion onboard the Bluefin 12 as a complementary sensor technology capable of localizing on buried objects. It is currently undergoing separate evaluation onboard a winged version of the Bluefin 12.

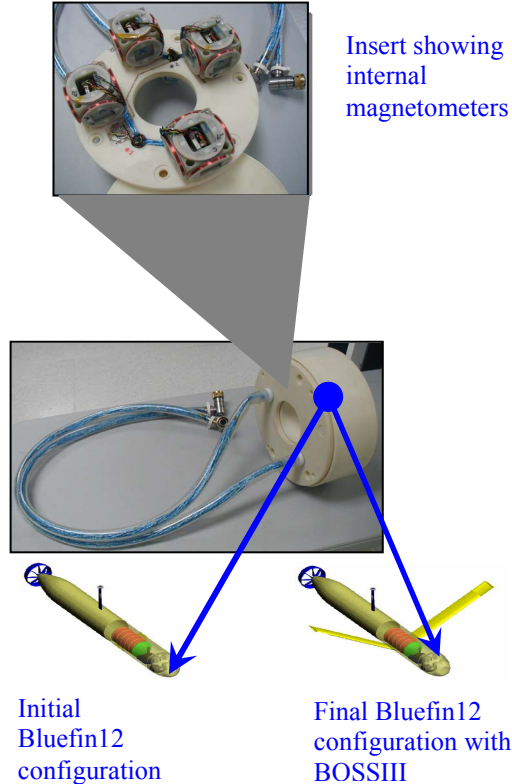


Fig. 1. The uuv-RTG sensor head with two Integration Options

II. RTG/BOSS FUSION FEASIBILITY DEMONSTRATED ON BOSS SLED

While the small AUVs and candidate sensors were being developed, The Naval Surface Warfare Center Panama City (NSWC PC), teamed with FAU, demonstrated the feasibility of using magnetic sensors fused with sonar to detect and locate buried mines and ordnance. This was accomplished by integrating an earlier version of the RTG, the RTG-fp, with FAU's BOSS II system, on a towed sled, Fig. 2, originally built by FAU for the BOSS II system. This sled was designed to have the high stability necessary for the sonar and a low magnetic signature, making it ideal for both technologies. Both the RTG-fp and the BOSSII are capable of providing single pass localization against buried and proud targets, allowing the data to be easily merged. After magnetic characterization, the RTG-fp and the BOSS II underwent several days of underwater evaluation using the BOSS sled on target sites in St. Andrews Bay near NSWC PC. The sled was towed by a large catamaran which was also selected for both stability and low magnetic signature. The tow configuration is shown in Fig. 3. Not visible in the photo was the deadweight depressor used to pull the sled below the surface. While underway, the depth of the sled in the water column was controlled by the amount of tow cable paid out from the catamaran.



Fig. 2. View of the BOSS sled with the RTG-fp



Fig. 3. Tow configuration of the RTG/BOSS sled combo at NSWC PC

Two significant sea tests provided “proof of feasibility” for the RTG technology and the Vaizer-Lathrop-Bono tracking algorithms [13] for the magnetic localization of buried ferrous targets. The test range for this evaluation consisted of various buried targets located in St. Andrews Bay at a depth of 10 meters.

The results of these tests are displayed in Fig. 4 and Fig. 5. The data shown in Fig. 4 were taken during adverse weather conditions and with all five gradients operating at moderate to high electronic noise levels caused by the lack of sync between the drive oscillators on the four magnetometers. Due to high winds during the test period, only a short segment of the range was transversed where the targets were in detection range.

Weather conditions were better during the second testing period and the entire test range could be covered, as shown in Fig 5. However, problems related to age and obsolescence of certain electronic components in the RTG-fp that was provided for that test, prevented the collection of large quantities of high quality data. Only two gradiometer channels were working with sensitivity better than $0.5 \text{ nT/m-Hz}^{1/2}$ at 0.1 Hz. This limited the localization algorithms to only two-channel localization instead of the five required for high-fidelity localization and moment classification. Instead the RTG operated as a simple coincidence detector,

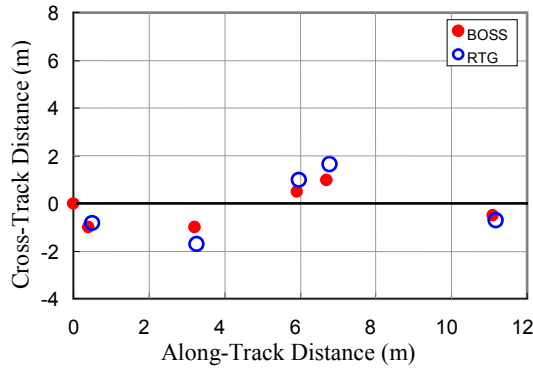


Fig. 4. Composite RTG-BOSS target map synthesized from Dec '03 Surveys

providing much weaker localization than obtained earlier. In Fig. 5, the positions of high-confidence magnetic contacts are displayed by red circular icons with a diameter of 5 feet, overlaid on a BOSS image of the target field. These results were synthesized from 3-19 kHz data collected with the BOSS sled at a nominal height of 2.5 meters above the sea floor. Burial for the targets ranged from 75% of target covered in the medium sand bottom to 1' below the sand water interface. Diver ground-truth target locations are represented by a small red dot at the center points of the dotted red circles.

The RTG localization circles are a composite of 14 runs over the target field. Good coincidence of along-track position for the acoustic and magnetic contacts was obtained with the uuv-RTG detecting all targets except the 2' cylinder. The considerable scatter in the estimated target locations verifies the need for additional channels to reduce the error circle for more accurate localization. There were several GPS-related issues adding to the errors. The GPS was mounted to the towing surface craft, not to the sled towed under water so that the RTG position was inferred under the assumption that the tow sled was moving in a line directly behind the surface craft, not taking into account lateral offsets of the sled. There was also the standard errors encountered in using GPS taken over a period of time.

In conclusion, we have shown that a two-channel sensor can be used for coincidence detection to enhance classification confidence for BOSS, but it lacks valuable localization data for the enhanced classification previously demonstrated with 5-channel sensors.

III. THE uuv-RTG INTEGRATION WITH BLUEFIN 12

This section addresses the mechanical and electrical aspects of the integration of the new uuv-RTG, manufactured by Quantum Magnetics [14] and the Bluefin 12. It also highlights the resulting data collected from this integration.

A. The New uuv-RTG

The new uuv-RTG system consists of a sensor head and an aluminum electronics bottle. These objects are separated by 140 mm. The motion capability of the uuv-

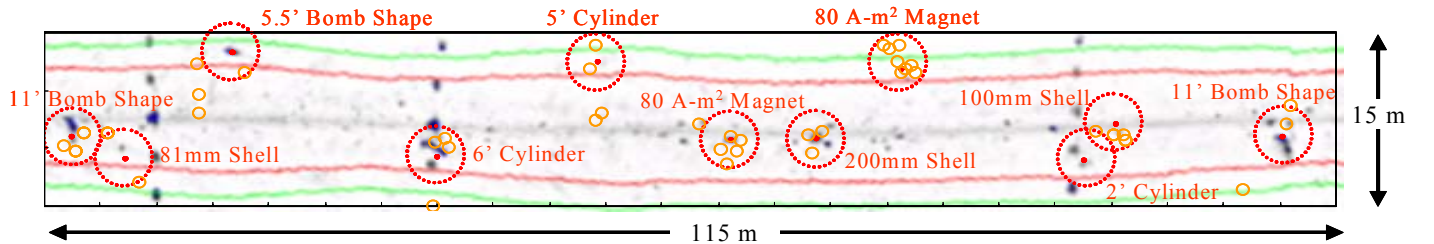


Fig. 5. Plot of sea data taken in May '04 with only three-of-five gradients operational. Orange circles are the localization error ranges of the RTG-fp caused by failed gradients. Black contacts are BOSS localizations

RTG, Fig. 1, is based on a three-sensor gradiometer concept developed by Dr. Roger Koch of IBM. The sensor head consists of four, 3-axis fluxgate magnetometers, each located within a 3-axis Helmholtz coil. Three of these sensors are separated from each other by a 165 mm baseline and are used to form six localization gradients, of which only five are independent. A fourth 3-axis sensor, is centrally located on the same base plate, and used as a reference sensor to provide magnetically related feedback to the 3 gradient generating sensors. Note that although this reference sensor has the three-axis Helmholtz coils, they are not connected. This allows the sensor to respond to the ambient magnetic variations, which produce the “nulling corrections” applied to the other three sensors that generate the localization gradients. This greatly reduces the motion response in the compensated sensors and allows them to operate in a nominal, low field environment.

The uuv-RTG, with appropriate software algorithms and navigational information, is capable of supplying range, bearing and magnetic moment data from a single pass. Complete details on the construction and theory of operation are presented in a companion paper [14]. The uuv-RTG excels in the localization of buried ferromagnetic objects. Because it is limited to ferromagnetic detection and has an inherent short range, it is considered only a part of the sensor suite. Other sensors, such as the BOSS III sonar, are necessary to fill in the localization gaps for nonferrous targets.

B. Bluefin 12

The Bluefin 12, shown in Fig. 7 with the nose removed to expose the uuv-RTG sensor head, is a wet (free-flooded) modular vehicle with a 324 mm outer diameter and a variable length dependent on the modular mission package. It is designed with a unique, articulating thruster on the stern for both propulsion and navigation control. Additional care has been taken to minimize the amount of EMI generated on the vehicle either by the construction material or electrically. Its modular design and absence of active control fins near the payload section in the nose make it an attractive platform for magnetic sensing since the mission package is isolated from the sources of EMI generated by vehicle operations.

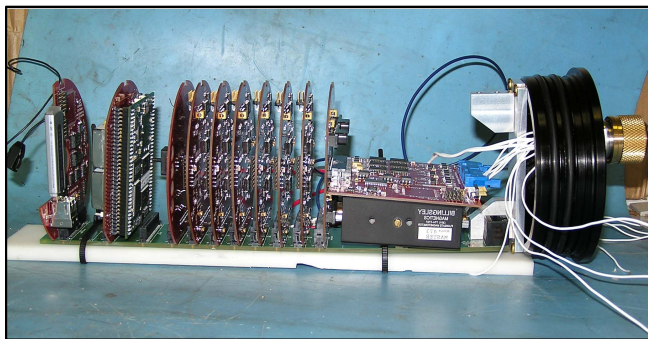


Fig. 6 uuv-RTG Electronics Stack

After initial testing in a stand-alone mode onboard a standard Bluefin 12, the uuv-RTG will be integrated with BOSS III into a winged version of Bluefin 12 where the wings are required to accommodate BOSS’s line array of acoustic receivers. This winged version is shown in Fig. 8.

C. Physical Integration

Referring to Fig. 9, the RTG is very tightly coupled to an inflexible baseplate, which is firmly attached to the Bluefin 12. The electronics bottle is located 483 mm away and mounted above the BOSS III bottle location, with two spaced cradles and caps. In addition, five auxiliary sensors are located throughout the vehicle for noise mitigation.

The sensor head is connected to the electronics by the two oil-filled, multi-conductor cables visible in Fig. 1. The



Fig. 7 Bluefin12 (Standard Version)



Fig. 8 Bluefin 12 (Winged Version)

electronics bottle is connected to the system through five underwater connectors located on the top cap; two for the sensor head, one for the auxiliary current sensors, and one for the auxiliary magnetometer's harness. The fifth connector connects the RTG electronics to the Bluefin 12. This host connection from the Bluefin 12 supplies both a 32-volt power connection and Ethernet communications. The Ethernet connection is used to supply navigational information as well as simple commands to the uuv-RTG, such as "turn on", "take data", "stop data" and "turn off."

Earlier analysis of the BOSS sled data confirmed that the sled was magnetically quiet. Only one magnetometer located near the BOSS transducer transformer provided any useful information for noise mitigation. While considerable attention was taken to make the Bluefin 12 as low mag as possible, some interference could not be avoided because of its small size and the close proximity of the EMI sources to the uuv-RTG. Isolating the uuv-RTG only minimized the effect. The EMI was determined to be from multiple sources such as the thruster, electrical transformers, and current loops, each requiring individual attention. For this application, a noise mitigation package was designed and implemented. It consists of three, 3-axis fluxgates magnetometers and two current sensors. Referring to Fig. 9, auxiliary noise mitigation magnetometers were placed near the BOSS transformer, the propulsion unit and the vehicle control electronics respectively. Current sensors, also used for noise mitigation, were placed in line with both the battery and thruster current leads.

IV. EXPERIMENTAL TESTS

After initial integration, the uuv-RTG/Bluefin 12 system was magnetically characterized using the NSWPC non-magnetic motion table facility shown in Fig. 10. This facility is located in a controlled environment and isolated from extraneous magnetic sources. The two-story building, shown in the background of Fig. 10, is constructed almost entirely of wood and plastic, with the exception of small aluminum or stainless steel fasteners. Fig. 11 shows the Bluefin 12 in place on a 3-axis nonmagnetic motion table located on the second floor of that building. The three drive motors, required for the three-axis motion, and shown in the foreground of Fig. 10, are separated from the table by 40 meters for magnetic isolation. They are coupled to the table by endless loops of Kevlar line, also seen in the figure. These drives can create single, dual or 3-axis motion, with each axis controlled individually for both amplitude and frequency. Single and dual axis motion is used for balancing the system, while 3-axis motion is used to simulate the at-sea conditions used for the final characterization. Experimental testing of the uuv-RTG integrated with the Bluefin 12 is just beginning and is still in the magnetic characterization stage.

At the time of writing, the RTG/Bluefin 12 combo was still undergoing tests on the motion table, as seen in Fig. 11. To determine its underwater capability, a series of progressive tests have been completed and data collected. Fig. 12a is a PSD of data taken with the RTG operating in place on the un-powered Bluefin 12. Fig. 12b is a PSD taken

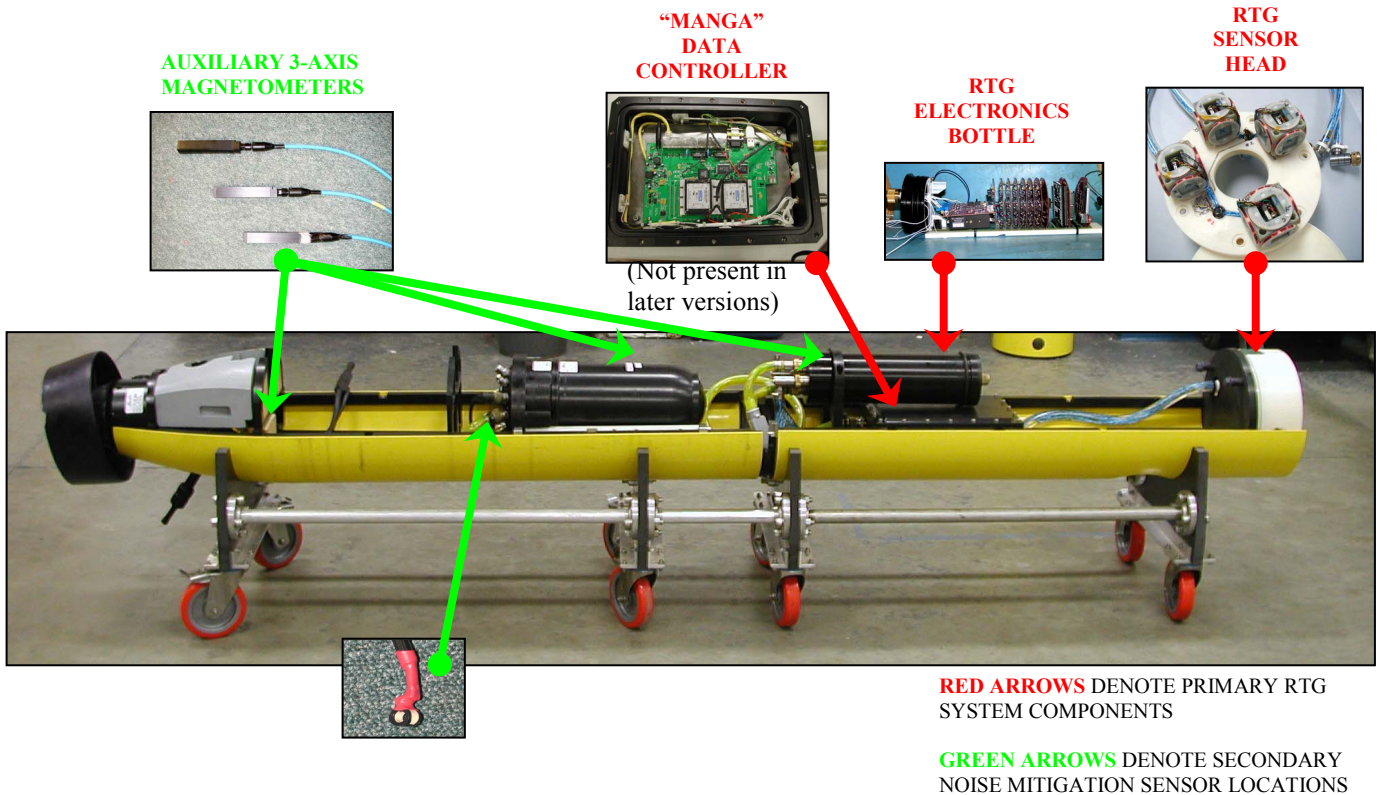


Fig. 9. The location of the uuv-RTG components onboard the Bluefin 12 (early prototype configuration)

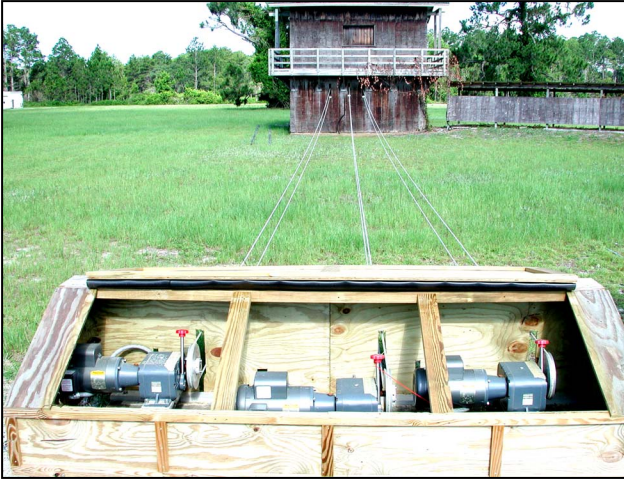


Fig. 10. View of nonmagnetic motion table facility showing drive motors

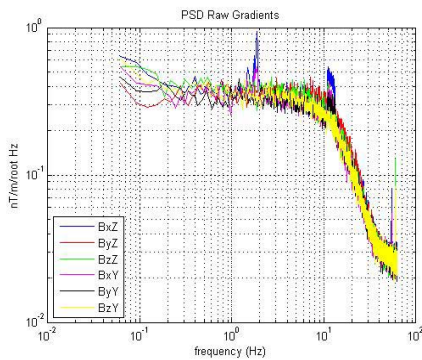


Fig. 11. Bluefin 12/RTG in place on CSS nonmagnetic motion table

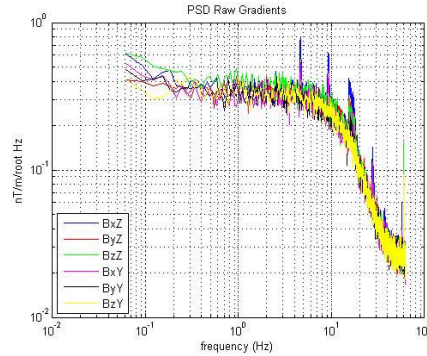
with both uuv-RTG and Bluefin 12 fully powered. Finally, Fig. 12c is the PSD with the Bluefin 12 powered and undergoing typical 3-axis simulated sea motion (± 1 degree and with periods between 3 and 10 seconds). With the AUV running around 4 Knots, the band of interest for target localization is somewhere around 0.1 Hz to 1 Hz. Using these two frequencies as reference points and comparing the three Power Density Spectra in Fig. 12, these early results indicate that both the EMI from Bluefin 12 and the motion generated noise in the band of interest, are at or below the noise level of the uuv-RTG operating alone. It should be noted that several of the Bluefin 12 systems did generate very narrow bands of detectable interference but they were above the frequency band of interest. As an example, Fig. 12b shows a sharp spike at 5 Hz with harmonics. Experimental analysis determined that the propulsion unit and motor on the Bluefin 12, running at 300 rpm (which is

5 Hz) produced this spike. It is not expected to be an issue since it is considerably above the band of interest and of small amplitude. Even though this EMI is anticipated to become larger when the propeller is loaded with seawater, it still is not expected to be a problem.

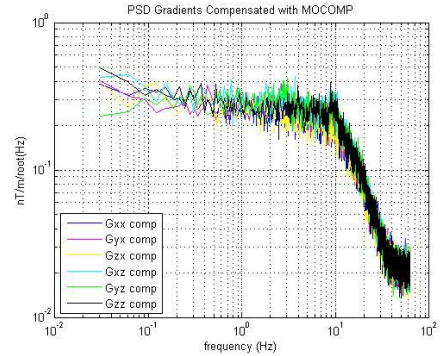
The PSD, as shown in Fig. 12c, is a simulation of the worst case scenario of the system operating at sea after the motion generated noise has been removed using a program called MOCOMP. MOCOMP uses a multiple referencing technique to selectively incorporate data from the noise mitigation sensors and the active uuv-RTG sensors as well. The spectrum in Fig. 12c showed no discernable increase in noise level either at 0.1 Hz or 1 Hz. Future tests will incorporate these early results. They will be compared with sea noise levels obtained once sea testing is instigated. These initial measurements indicate that the uuv-RTG will be able to operate near or at sensor noise when operating at sea.



(a.)



(b.)



(c.)

Fig. 12. Power Spectral Density Plots of the uuv-RTG gradients taken onboard the Bluefin 12 under conditions of (a.) uuv-RTG alone powered, (b.) both uuv-RTG and Bluefin 12 fully powered, and (c.) entire system powered and in simulated sea motion.

The initial work was not done in real-time. This post-processing of the data requires complex, time-consuming algorithms. When processing data in real-time, the algorithms must be simplified. This will limit the data quality, directly affecting its accuracy. While the initial data is definitely encouraging, the same results, as demonstrated here, will be affected by the time available for processing.

V. CONCLUSION

In conclusion, uuv-RTG integrated with the Bluefin 12 is capable of detecting and localizing ferromagnetic targets in an undersea environment. Near sensor noise levels are achieved using specialized algorithms and a noise mitigation package. Together, they reject platform generated EMI and motion noise. Initial sensor fusion has been accomplished between the RTG and the Boss technology providing enhanced capability for underwater mine localization and classification.

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